



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Final Presentation
Team 56: OpenROAD VLSI Design 2
Kevin, Talha, Githin, & Giovanni
TA: Michael Johnson-Moore
Sponsor: Kevin Nowka

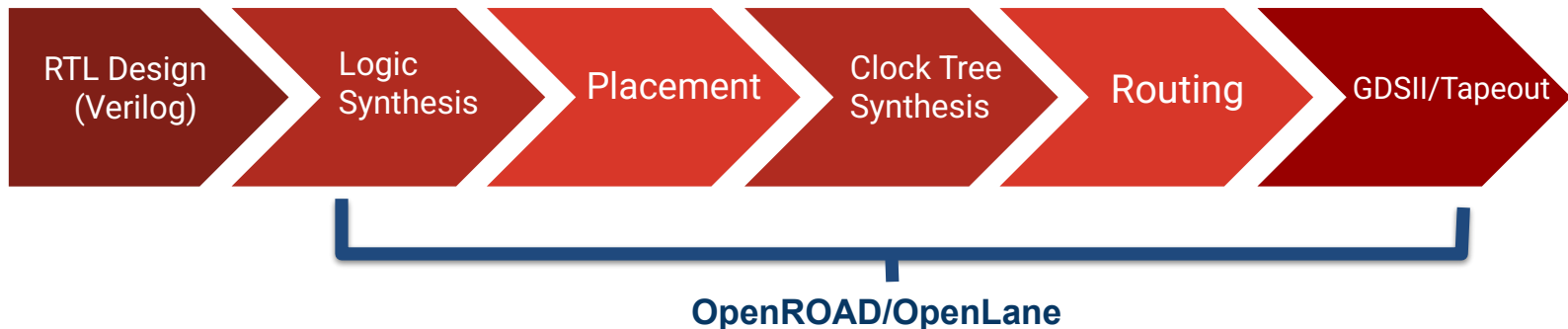
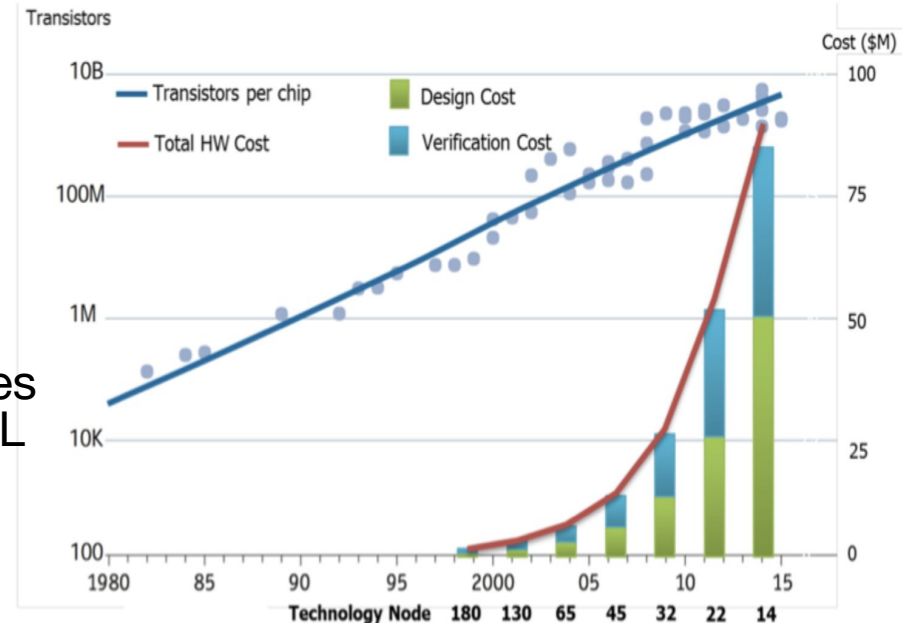
Problem Overview

Problem statement:

- Traditional RTL-to-GDSII chip design is expensive, creating a cost barrier to developing a System on a Chip (SoC).

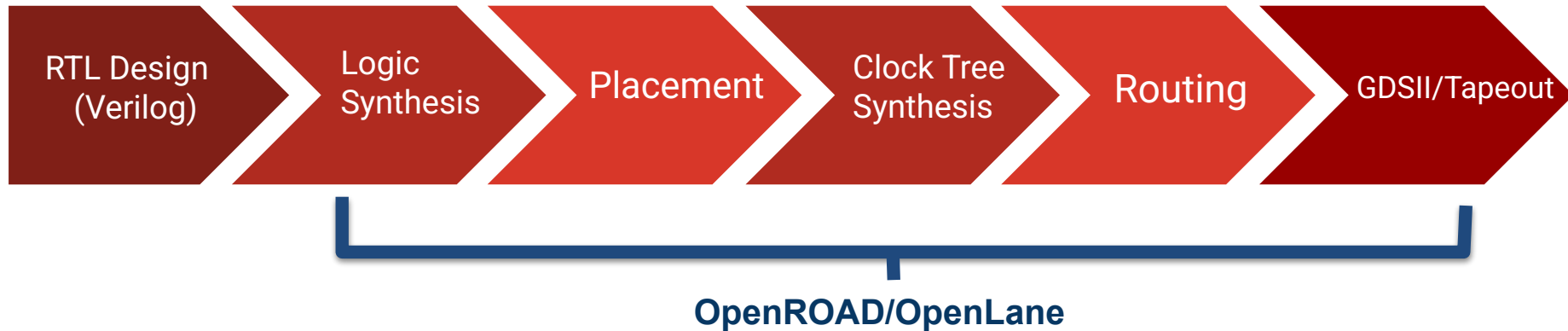
Our Solution:

- Create a 16-bit RISC-V using OpenROAD/OpenLane, which automates chip design by only needing only an RTL description to generate GDSII, significantly reducing cost.

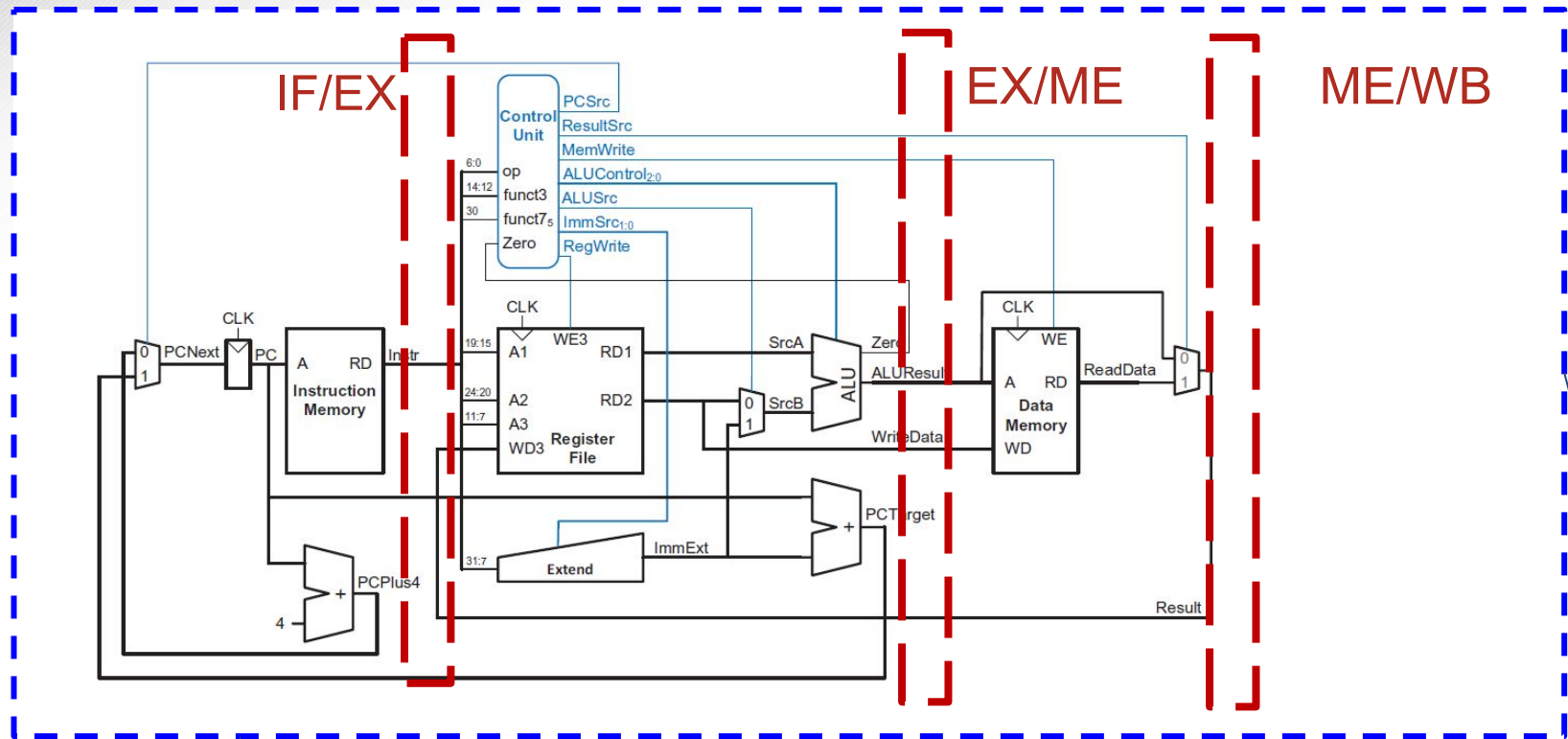




Design Flow



Integrated Project Diagram



General Purpose
Input Pins

General General
Output Pins

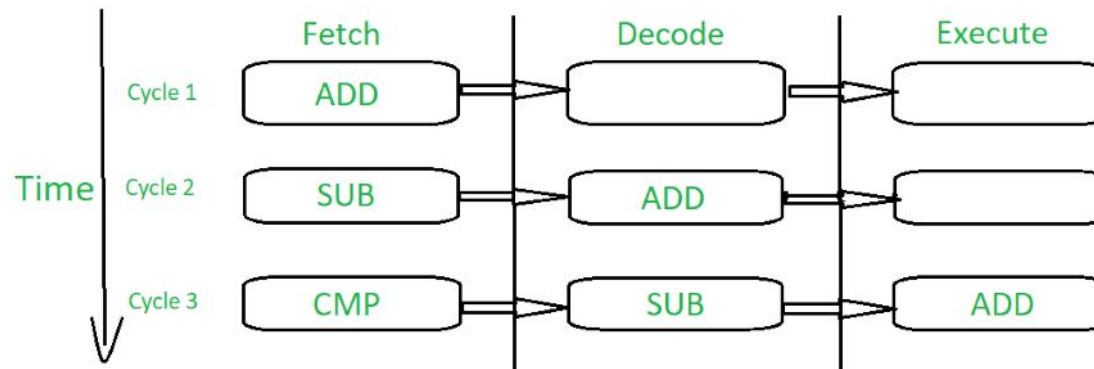
Processor & Pipeline Specifications

Processor

- Scaled down industry standard 32-bit RISC-V ISA down to 16-bit

Pipeline

- Allows for our CPU to run three instructions at the same time
- Features three pipeline registers
- Handles data, load, and control hazards



Pipelined Processor Hurdles & Solutions

Hurdles

- **ISA Design**
 - Reducing to a 16-bit design limits operation space for instructions
- **Hazard Handling**
 - Accounting for data hazards where a second instruction needs data the first instruction hasn't wrote back yet
 - Accounting for control hazards where after we branch the CPU begins executing the wrong instruction

Solutions

- **ISA Design**
 - Reallocate bits to prioritize critical operations
- **Hazard Handling**
 - Implemented forwarding unit for data hazards to directly forward the data from the pipelined register.
 - Implemented flush logic to flush the pipelined registers contents for the irrelevant instruction



Processor & Pipeline Results

```
=== -- Check I-types (Positive)-- ===
```

```
PASS | x0 = 0
PASS | ADDI x1 = 2
PASS | ADDI x2 = 3
PASS | XORI x3 = 1
PASS | XORI x4 = 2
PASS | ORI x5 = 1
PASS | ORI x6 = 3
PASS | ANDI x7 = 0
```

```
=== -- Check I-types (Negative) -- ===
```

```
PASS | x0 = 0
PASS | ADDI x1 = -1
PASS | ADDI x2 = -4
PASS | XORI x3 = -2
PASS | XORI x4 = -3
PASS | ORI x5 = -1
PASS | ORI x6 = -2
PASS | ANDI x7 = 0
```

```
-- Check R-type Arithmetic (use of forwarding)--
```

```
PASS | x0 = 0
PASS | ADD x1 = -2
PASS | SLT x2 = 0
PASS | SLTU x3 = 0
PASS | OR x4 = -2
PASS | SLL x5 = -32768
PASS | SUB x6 = -2
PASS | SRL x7 = 3
```

```
=== -- Check Store/Load (use of stalling) -- ===
```

```
PASS | x0 = 0                                got=0x0000
PASS | LW x4 = 3                             got=0x0003
PASS | LW x3 = -32768                        got=0x8000
PASS | LW x6 = -2                             got=0xfffe
```

```
=== -- Check Branch -- ===
```

```
PASS | x1 = 2, (flushes proceeding ADD -1)    got=0x0002
```

```
765000 | PC=0090 instr=020d ALU=0008 | x0=0000 x1=0002 x2=0002 x3=
775000 | PC=0000 instr=0000 ALU=0000 | x0=0000 x1=0002 x2=0002 x3=
785000 | PC=00a0 instr=0000 ALU=0000 | x0=0000 x1=0002 x2=0002 x3=
795000 | PC=00a2 instr=008f ALU=fffe | x0=0000 x1=0092 x2=0002 x3=
```

```
=== -- Check JAL -- ===
```

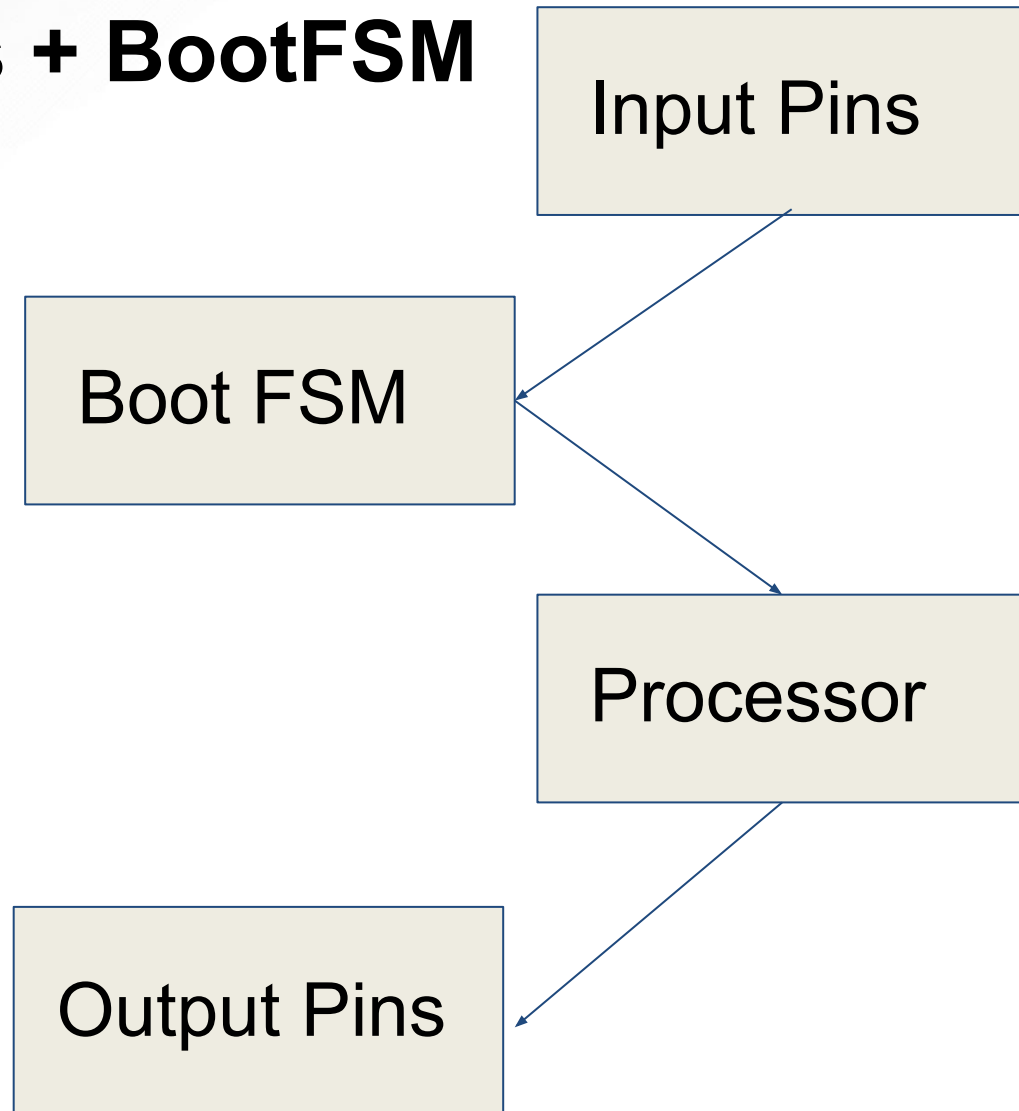
```
PASS | x1 = 0x0092                            got=0x0092
PASS | x2 = 2                                 got=0x0002
```

```
=====
Results: 32 PASSED, 0 FAILED
=====
```

```
ALL TESTS PASSED
```

I/O Pins + BootFSM

- Allows user to load program through input pins before execution
- 14-bit instruction transferred in 2 transfers (T1 + T2)
- Boot FSM resets to IDLE between transfers to prevent corruption



I/O Pins + BootFSM Hurdles & Solutions

Hurdles

- **Tiny Tapeout:**
 - Loading 16-bit instructions given pin limitations
 - Implementing Finite State Machine (FSM) for loading RAM addresses
- **OpenLane:**
 - Reducing Antenna Errors
 - Optimizing area utilization

Solutions

- **Tiny Tapeout:**
 - Load instructions into RAM using 2 clock cycles
 - FSM has 4 stages (IDLE, GOT_T1, Write, ACK)
- **OpenLane:**
 - Included diode insertion strategy for reducing antenna errors
 - Tested different values for core utilization, cell padding, etc.



I/O & FSM Results

PHASE 1: LOAD PROGRAM THROUGH PINS

```
Loading instr 0: ADDI x1, x0, 3 = 0x6081
Loading instr 1: ADDI x2, x0, 2 = 0x4101
Loading instr 2: ADD x0, x1, x2 = 0x2800
Loading instr 3: SUB x4, x1, x2 = 0x2A08
Loading instr 4: ADD x6, x1, x2 = 0x2B00
Loading instr 5: OR x3, x1, x2 = 0x2998
Loading instr 6: XOR x2, x1, x2 = 0x2910
Loading instr 6: NOP = 0x0000
Loading instr 7: NOP = 0x0000
```

IMEM check:

```
memory[0]=0x6081 (expect 0x6081)
memory[1]=0x4101 (expect 0x4101)
memory[2]=0x2800 (expect 0x2800)
memory[3]=0x2a08 (expect 0x2A08)
memory[4]=0x2b00 (expect 0x2B00)
memory[5]=0x2998 (expect 0x2998)
memory[6]=0x2910
```

Final register snapshot:

```
x1 = 0x0003 (expect 0x0003)
x2 = 0x0001 (expect 0x0002)
x3 = 0x0003 (expect 0x0003)
x4 = 0x0001 (expect 0x0001)
x6 = 0x0005 (expect 0x0005)
```

```
FINAL: pass=6 fail=0
```

PHASE 3: EXECUTION TRACE THROUGH PINS

PC=0x0000	INSTR=0x6081	ALU=0x0003	x1=0x0000	x2=0x0000	x3=0x0000	x4=0x0000	x6=0x0000
PC=0x0002	INSTR=0x4101	ALU=0x0002	x1=0x0000	x2=0x0000	x3=0x0000	x4=0x0000	x6=0x0000
PC=0x0004	INSTR=0x2800	ALU=0x0005	x1=0x0000	x2=0x0000	x3=0x0000	x4=0x0000	x6=0x0000
PC=0x0006	INSTR=0x2a08	ALU=0x0001	x1=0x0003	x2=0x0000	x3=0x0000	x4=0x0000	x6=0x0000
PC=0x0008	INSTR=0x2b00	ALU=0x0005	x1=0x0003	x2=0x0002	x3=0x0000	x4=0x0000	x6=0x0000
PC=0x000a	INSTR=0x2998	ALU=0x0003	x1=0x0003	x2=0x0002	x3=0x0000	x4=0x0000	x6=0x0000
PC=0x000c	INSTR=0x2910	ALU=0x0001	x1=0x0003	x2=0x0002	x3=0x0000	x4=0x0001	x6=0x0000
PC=0x000e	INSTR=0x0000	ALU=0x0000	x1=0x0003	x2=0x0002	x3=0x0000	x4=0x0001	x6=0x0005

OpenLane Video

```

[[nix-shell:~]$
[[nix-shell:~]$
[[nix-shell:~]$
[[nix-shell:~]$
[[nix-shell:~]$
[[nix-shell:~]$ openlane ~/my_designs/CPU/config.json
[09:54:52] INFO      Loading the design configuration file has      config.py:716
                    generated the following warnings:
[09:54:52] WARNING  The DIODE_INSERTION_STRATEGY variable has been config.py:720
                    deprecated. See 'Migrating
                    DIODE_INSERTION_STRATEGY' in the docs for more
                    info.
[09:54:52] WARNING  The configuration variable                      config.py:720
                    'GRT_REPAIR_ANTENNAS' is deprecated. Please
                    check the docs for the usage on the
                    replacement variable 'RUN_ANTENNA_REPAIR'.
[09:54:52] INFO      Starting a new run of the 'Classic' flow with    flow.py:628
                    the tag 'RUN_2026-04-13_09-54-52'.
[09:54:52] INFO      Starting...                                      sequential.py:294

                    ----- Verilator Lint -----
[09:54:52] VERBOSE  Running 'Verilator.Lint' at                      step.py:1122
                    'my_designs/CPU/runs/RUN_2026-04-13_09-54-52/01
                    -verilator-lint'...

Classic - Stage 1 - Verilator Lint ----- 0/78 0:00:00

```




Gate Level Test

```
=====
PHASE 1: LOAD PROGRAM THROUGH PINS
=====
Loading instr 0: ADDI x1, x0, 3 = 0x6081
[ACK] received after 3 cycles
Loading instr 1: ADDI x2, x0, 2 = 0x4101
[ACK] received after 3 cycles
Loading instr 2: ADD x0, x1, x2 = 0x2800
[ACK] received after 3 cycles
Loading instr 3: SUB x4, x1, x2 = 0x2A08
[ACK] received after 3 cycles
Loading instr 4: ADD x6, x1, x2 = 0x2B00
[ACK] received after 3 cycles
Loading instr 5: OR x3, x1, x2 = 0x2998
[ACK] received after 3 cycles
Loading instr 6: XOR x2, x1, x2 = 0x2910
[ACK] received after 3 cycles
Loading instr 7: NOP = 0x0000
[ACK] received after 3 cycles
Loading instr 8: NOP = 0x0000
[ACK] received after 3 cycles
PASS | uio_oe=0x00 during LOAD (bidir=inputs)

=====
PHASE 2: RESET AND START EXECUTION
=====
PASS | uio_oe=0xFF during RUN (bidir=outputs)
PASS | PC=0 during reset via pin | got 0x0000

=====
PHASE 3: EXECUTION TRACE THROUGH PINS
=====
PC=0x0000 | INSTR=0x6081 | ALU=0x0003 | x1=0x0000 | x2=0x0000 | x3=0x0000 | x4=0x0000 | x6=0x0000
PC=0x0002 | INSTR=0x4101 | ALU=0x0002 | x1=0x0000 | x2=0x0000 | x3=0x0000 | x4=0x0000 | x6=0x0000
PC=0x0004 | INSTR=0x2800 | ALU=0x0005 | x1=0x0000 | x2=0x0000 | x3=0x0000 | x4=0x0000 | x6=0x0000
PC=0x0006 | INSTR=0x2a08 | ALU=0x0001 | x1=0x0003 | x2=0x0000 | x3=0x0000 | x4=0x0000 | x6=0x0000
PC=0x0008 | INSTR=0x2b00 | ALU=0x0005 | x1=0x0003 | x2=0x0002 | x3=0x0000 | x4=0x0000 | x6=0x0000
PC=0x000a | INSTR=0x2998 | ALU=0x0003 | x1=0x0003 | x2=0x0002 | x3=0x0000 | x4=0x0000 | x6=0x0000
PC=0x000c | INSTR=0x2910 | ALU=0x0001 | x1=0x0003 | x2=0x0002 | x3=0x0000 | x4=0x0001 | x6=0x0000
PC=0x000e | INSTR=0x0000 | ALU=0x0000 | x1=0x0003 | x2=0x0002 | x3=0x0000 | x4=0x0001 | x6=0x0005
PC=0x0010 | INSTR=0x0000 | ALU=0x0000 | x1=0x0003 | x2=0x0002 | x3=0x0003 | x4=0x0001 | x6=0x0005
PC=0x0012 | INSTR=0xxxxx | ALU=0xxxxx | x1=0x0003 | x2=0x0001 | x3=0x0003 | x4=0x0001 | x6=0x0005
PC=0x00XX | INSTR=0xxxxx | ALU=0xxxxx | x1=0x0003 | x2=0x0001 | x3=0x0003 | x4=0x0001 | x6=0x0005
PC=0xxxxx | INSTR=0xxxxx | ALU=0xxxxx | x1=0x0003 | x2=0x0001 | x3=0x0003 | x4=0x0001 | x6=0x0005

=====
PHASE 4: CHECK ALL REGISTER VALUES THROUGH PINS
=====
PASS | x1=3 after ADDI x1,x0,3 | got 0x0003
PASS | x2=1 after XOR x2,x1,x2 (3^2=1) | got 0x0001
PASS | x3=3 after OR x3,x1,x2 (3|2=3) | got 0x0003
PASS | x4=1 after SUB x4,x1,x2 (3-2=1) | got 0x0001
PASS | x6=5 after ADD x6,x1,x2 (3+2=5) | got 0x0005

=====
FINAL: pass=6 fail=0
```

Area Spec

```

26 Updating metrics...
27 Cell type report:
28   Fill cell          Count      Area
29   Tap cell           1024      3843.69
30   Antenna cell       27756     34728.31
31   Buffer              22137     55395.63
32   Buffer              8         30.03
33   Clock buffer        1120     14821.72
34   Timing Repair Buffer 6504     59583.40
35   Inverter            34         135.13
36   Clock inverter      462      6602.58
37   Sequential cell     4503     91163.68
38   Multi-Input combinational cell 9139     94677.05
39   Total               72687    360981.21
40
41 Writing OpenROAD database to '/Users/talhaibrahim/my_designs/CPU/run
110   "design_core_area": 1941700,
111   "design_instance_count_stdcell": 71663,
112   "design_instance_area_stdcell": 357138,
113   "design_instance_count_macros": 0,
114   "design_instance_area_macros": 0,
115   ..... "design_instance_utilization": 0.18393,
116   "design_instance_utilization_stdcell": 0.18393,
117   "design_instance_count_class:buffer": 8,
118   "design_instance_count_class:inverter": 34

```




Die Area

```
26 [INFO IFP-0001] Added 512 rows of 3031 site unitld.  
27 [INFO IFP-0030] Inserted 0 tiecells using sky130_fd_sc_hd__conb_1/L0.  
28 [INFO IFP-0030] Inserted 0 tiecells using sky130_fd_sc_hd__conb_1/HI.  
29 [INFO] Extracting DIE_AREA and CORE_AREA from the floorplan  
30 [INFO] Floorplanned on a die area of 0.0 0.0 1405.36 1416.08 (μm).  
31 [INFO] Floorplanned on a core area of 5.52 10.88 1399.78 1403.52 (μm).  
32 Writing metric design__die__bbox: 0.0 0.0 1405.36 1416.08  
33 Writing metric design__core__bbox: 5.52 10.88 1399.78 1403.52  
34 Setting global connections for newly added cells...
```



STA

Fanout	Cap	Slew	Delay	Time	Description
		0.150000	0.000000	0.000000	clock clk (rise edge)
			0.000000	0.000000	clock network delay (ideal)
			8.000000	8.000000	^ input external delay
140	0.506271	1.698526	1.184126	9.184126	^ rst_n (in)
					rst_n (net)
		1.698526	0.000000	9.184126	^ _20475_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
				9.184126	data arrival time
		0.150000	40.000000	40.000000	clock clk (rise edge)
			0.000000	40.000000	clock network delay (ideal)
			-0.250000	39.750000	clock uncertainty
			0.000000	39.750000	clock reconvergence pessimism
				39.750000	^ _20475_/CLK (sky130_fd_sc_hd__dfrtp_2)
			-0.308231	39.441769	library recovery time
				39.441769	data required time
				39.441769	data required time
				-9.184126	data arrival time
				30.257647	slack (MET)



DRC

```
263      390000 uses
264      395000 uses
265      400000 uses
266      405000 uses
267      410000 uses
268      415000 uses
269 [INFO] Loading tt_um_riscv16
270
271 DRC style is now "drc(full)"
272 Loading DRC CIF style.
273 No errors found.
274 [INFO] COUNT: 0
275 [INFO] Should be divided by 3 or 4
276 [INFO] DRC Checking DONE (/Users/talhaibrahim/my_designs/CPU/runs/RUN_2026-03-30_08-34-19/62-magic-drc/reports/drc_vio
277 [INFO] Saving mag view with DRC errors (/Users/talhaibrahim/my_designs/CPU/runs/RUN_2026-03-30_08-34-19/62-magic-drc/v
278 [INFO] Saved
279
```



LVS

```
4225      uio_in[0]          |uio_in[0]
4226      uio_in[2]          |uio_in[2]
4227      uio_in[7]          |uio_in[7]
4228      clk                 |clk
4229      VGND                |VGND
4230      VPWR                |VPWR
4231      ena                 |ena
4232      -----
4233      Cell pin lists are equivalent.
4234      Device classes tt_um_riscv16 and tt_um_riscv16 are equivalent.
4235
4236      Final result: Circuits match uniquely.
4237      .
4238
```




Power

CPU > runs > RUN_2026-03-30_09-00-53 > 12-openroad-staprepr > nom_tt_025C_1v80 > ≡ power.rpt

report_power

nom_tt_025C_1v80 Corner

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)
Sequential	4.690573e-03	3.095727e-04	3.896356e-08	5.000184e-03 39.6%
Combinational	5.477844e-03	2.148002e-03	4.657001e-08	7.625893e-03 60.4%
Clock	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00 0.0%
Macro	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00 0.0%
Pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00 0.0%
Total	1.016921e-02	2.457575e-03	8.553488e-08	1.262687e-02 100.0%
	80.5%	19.5%	0.0%	

Temp Sweep

```
./sweep.sh
VCD info: dumpfile ./tb/waveform/top.vcd opened for output.
Corner: tt
PASS x1 ADDI x1,x0,3      = 0x0003
PASS x2 XOR  x2,x1,x2     = 0x0001
PASS x3 OR   x3,x1,x2     = 0x0003
PASS x4 SUB  x4,x1,x2     = 0x0001
PASS x6 ADD  x6,x1,x2     = 0x0005
5/5 passed
top_tb_gl.v:127: $finish called at 1438000 (1ps)
VCD info: dumpfile ./tb/waveform/top.vcd opened for output.
Corner: ss
PASS x1 ADDI x1,x0,3      = 0x0003
PASS x2 XOR  x2,x1,x2     = 0x0001
PASS x3 OR   x3,x1,x2     = 0x0003
PASS x4 SUB  x4,x1,x2     = 0x0001
PASS x6 ADD  x6,x1,x2     = 0x0005
5/5 passed
top_tb_gl.v:127: $finish called at 1438000 (1ps)
VCD info: dumpfile ./tb/waveform/top.vcd opened for output.
Corner: ff
PASS x1 ADDI x1,x0,3      = 0x0003
PASS x2 XOR  x2,x1,x2     = 0x0001
PASS x3 OR   x3,x1,x2     = 0x0003
PASS x4 SUB  x4,x1,x2     = 0x0001
PASS x6 ADD  x6,x1,x2     = 0x0005
5/5 passed
top_tb_gl.v:127: $finish called at 1438000 (1ps)
```

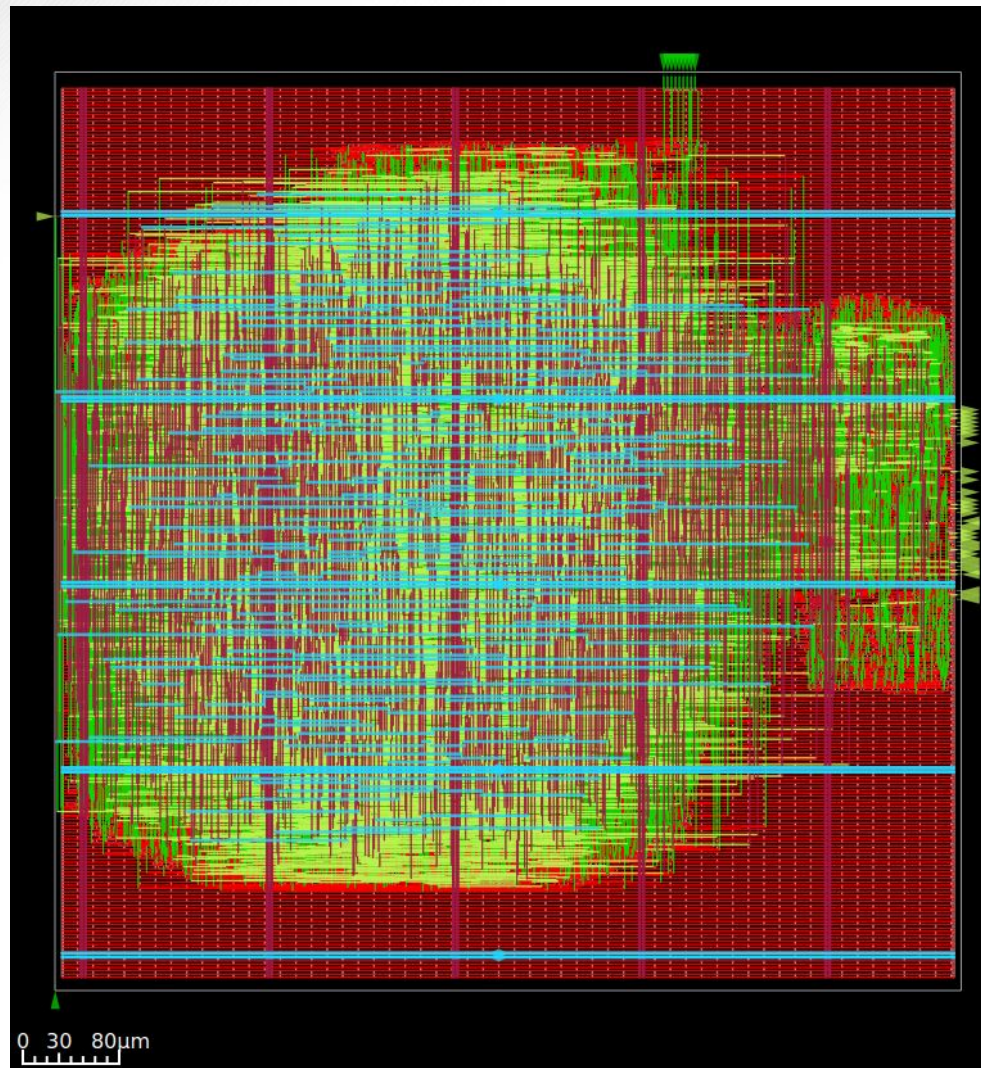


Subsystem Specs/Test Results

	Requirements	Results
Tile Size	2 tile	1.96 tile
DRC	Pass	Pass
Operating Temperature Range	-40C to 125C	Pass
LVS	Pass	Pass
Primary Input Power	200 μ W	12.62 mW
Boot/Rest/Initialization	Pass	Pass
Clock Interface	10 MHz	25 Mhz



Final GDSII Layout





Conclusions

- Final System Design Changes
 - Modified Instruction Set Architecture to compute larger values
 - Changed from 5-stage to 3-stage pipeline
 - Implemented pins using TinyTapeout format to interface with real world
- Current Status
 - The major instruction types and pipeline are fully tested and validated, we also are working on optional minor tests in progress - to be complete in 1 week
 - Successfully ran processor through OpenLane
 - Generated Final GDSII output from OpenLane to match design requirements



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Questions?